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Course: Software Testing and Quality Assurance Laboratory	Code: BIT26PE11
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Assignment 10: Application of Ishikawa's Quality Tools – Fishbone Diagram and Pareto Analysis	

1. OBJECTIVE

- To study and apply **Ishikawa's 7 Quality Tools** in software testing.
- To identify root causes of defects in the **Hotel Management System**.
- To construct a **Fishbone Diagram** for cause-effect analysis.
- To perform **Pareto Analysis** to prioritize major defects.
- To improve system quality by focusing on critical issues.

2. INTRODUCTION

Ishikawa's Quality Tools are basic techniques used for **quality control and problem-solving** in software systems.

Seven Quality Tools:

1. Cause and Effect Diagram (Fishbone Diagram)
2. Pareto Chart
3. Check Sheet
4. Histogram
5. Control Chart
6. Scatter Diagram
7. Flow Chart

In our **Hotel Management System project**, these tools help analyze issues found during:

- GUI Testing
- Performance Testing
- Functional Testing

(as performed in previous assignments)

3. SCENARIO

During testing of the **Hotel Management System**, multiple issues were observed in:

- Room booking process
- User login system
- UI responsiveness
- Payment module

These issues were identified from previous testing activities such as GUI and performance testing

4. DATASET

Identified Issues and Frequency

Issue	Frequency
Room not clean / UI mismatch	35
Slow system response	25
Booking errors	15
Login failures	10
Payment issues	8
Other minor bugs	7

5. FISHBONE DIAGRAM (CAUSE-EFFECT ANALYSIS)

Problem: Frequent Defects in Hotel Management System

Fishbone Diagram (6M Categories)

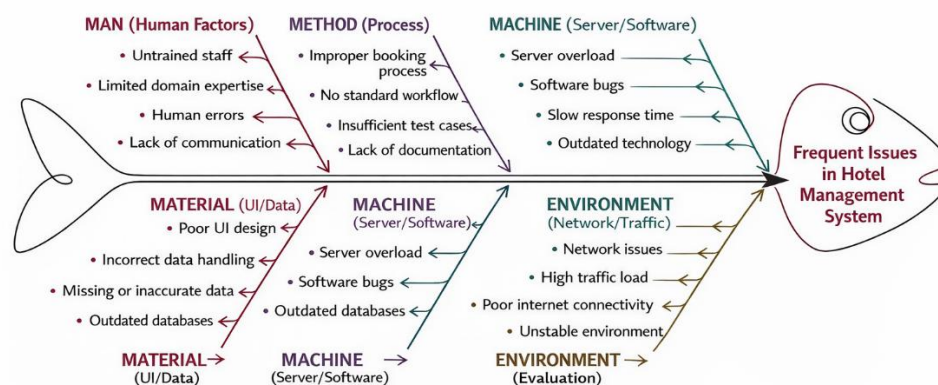


Fig 1: Fishbone Diagram (Cause-Effect Analysis)

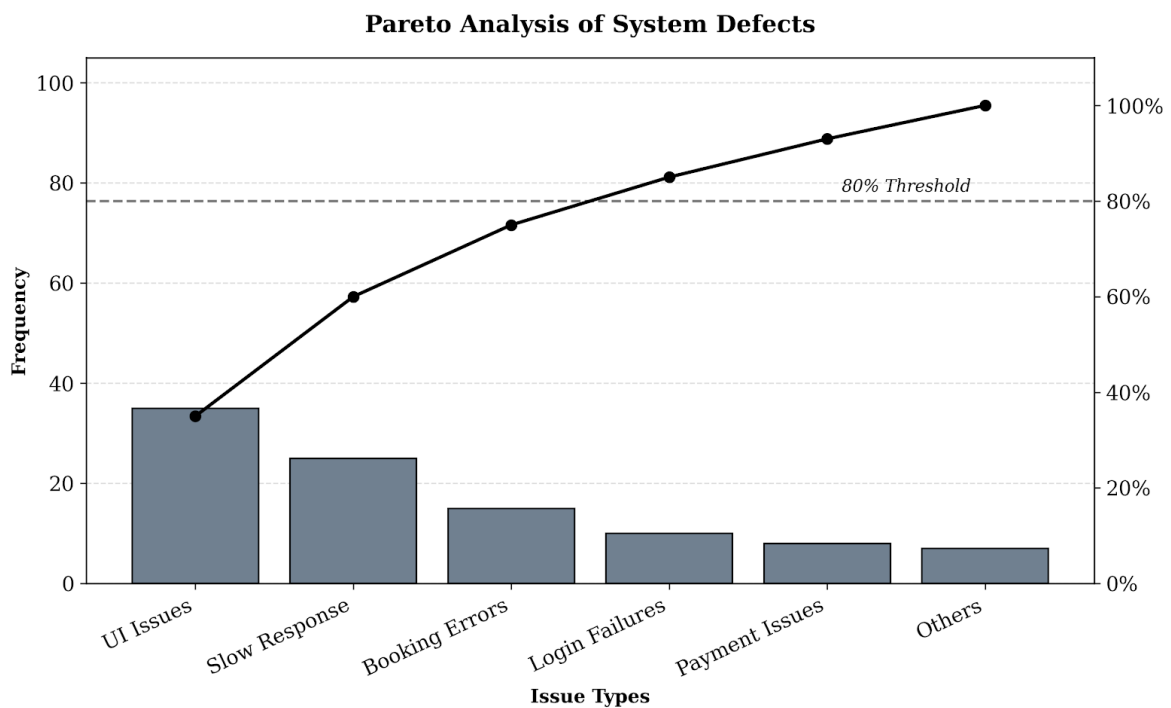
Explanation of Causes:

- **Man:** Lack of training → incorrect data entry and poor handling
- **Machine:** Server overload and bugs → slow response and crashes
- **Method:** Improper booking process → errors in workflow
- **Material:** Poor UI design → confusion and incorrect usage
- **Measurement:** No tracking → issues not identified early
- **Environment:** Network/server issues → system delay

6. PARETO ANALYSIS

Step 1: Sorted Data

Issue	Frequency	Cumulative Frequency	Percentage
Room/UI Issues	35	35	35%
Slow Response	25	60	60%
Booking Errors	15	75	75%
Login Failures	10	85	85%
Payment Issues	8	93	93%
Others	7	100	100%



Step 2: Analysis

- Top issues contributing to defects:
 - Room/UI Issues
 - Slow Response
 - Booking Errors

These contribute to **~75–80% of total problems**

Pareto Principle (80/20 Rule):

- **80% of problems are caused by 20% of factors**
- Focus should be on:
 - UI improvements
 - Performance optimization
 - Booking system fixes

7. IMPLEMENTATION STEPS

Fishbone Diagram Steps:

1. Identify the main problem
2. Categorize causes using 6M
3. List all possible causes
4. Analyze root causes

Pareto Analysis Steps:

1. Collect defect data
2. Arrange in descending order
3. Calculate cumulative frequency
4. Identify major contributors

8. RESULT AND ANALYSIS

- Fishbone diagram identified **root causes across system layers**
- Pareto analysis showed:
 - Majority issues come from **UI, performance, and booking logic**
- Based on this:
 - Focused testing and fixes can improve system quality significantly

9. CONCLUSION

The application of Ishikawa's Quality Tools helped in:

- Identifying root causes using Fishbone Diagram
- Prioritizing critical issues using Pareto Analysis

This approach improves:

- System reliability
- User experience
- Testing efficiency

Thus, the **Hotel Management System becomes more stable and optimized.**